

Standard Operating Procedure: Bearing Inspection

SOP ID: SOP-BRG-02 | Revision: 1.4

1. Purpose

To outline the systematic procedure for inspecting rolling element bearings in industrial electric motors to detect early-stage degradation.

2. Safety Prerequisites

WARNING: Ensure the machine is completely de-energized and locked out (LOTO) if physical contact with the shaft or bearing housing is required.

3. Inspection Steps

- **Acoustic Evaluation:** Listen to the bearing during coast-down. Grinding, clicking, or whining noises indicate raceway spalling or lack of lubrication.
- **Thermal Check:** Use an IR thermometer. Bearing housing temperatures exceeding 80°C (176°F) indicate abnormal friction.
- **Vibration Analysis:** Attach an accelerometer to the bearing housing (horizontal and vertical planes). Record the velocity RMS (mm/s).
- **Visual Inspection:** Check the shaft seal for purged grease or discoloration. Darkened grease indicates severe overheating or contamination.